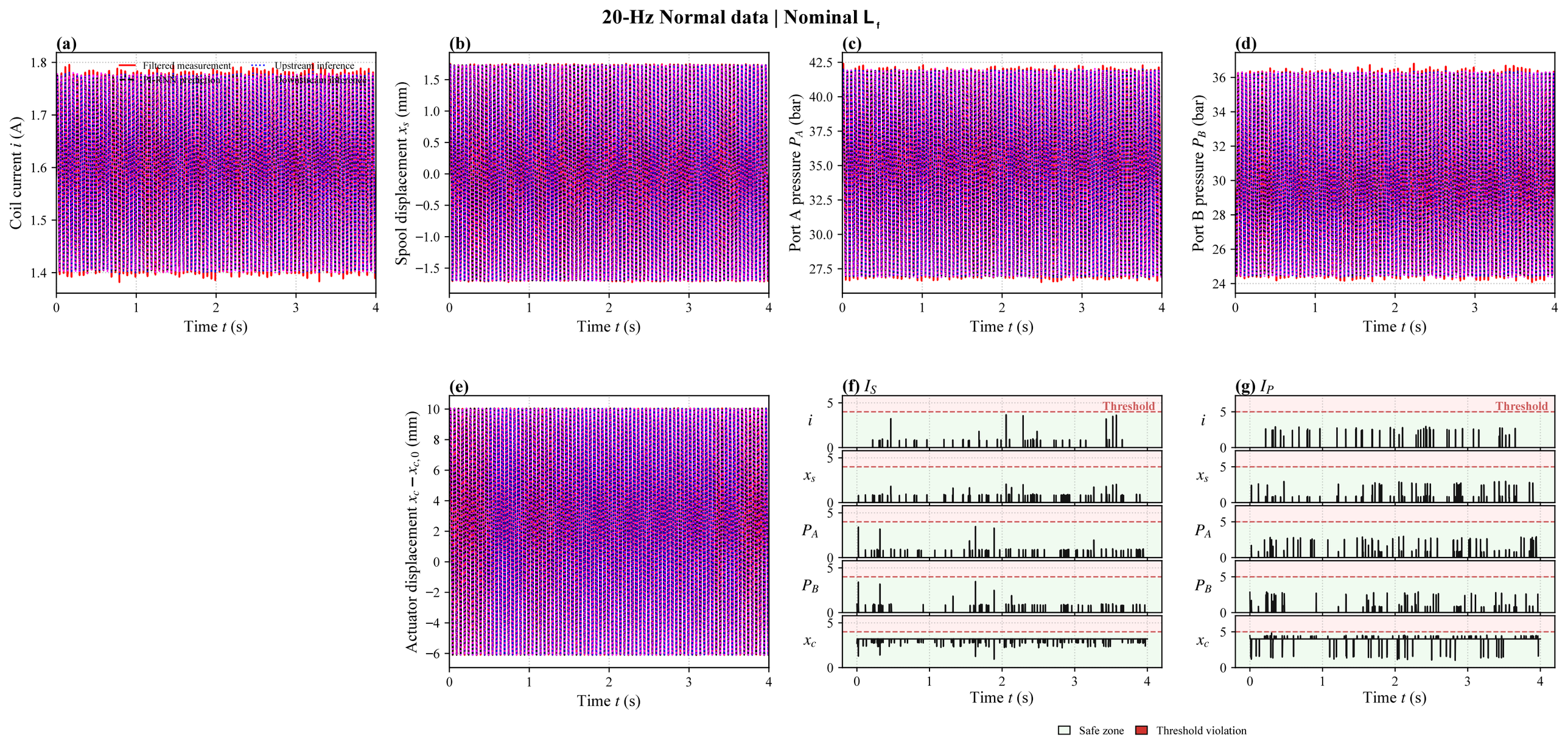

20-Hz Normal data | Nominal L_f
(a)
Filtered measurement
Upstream inference
PI-RNN prediction
Downstream inference
Coil current i (A)
Time t (s)
(b)
Spool displacement x_s (mm)
Time t (s)
(c)
Port A pressure P_A (bar)
Time t (s)
(d)
Port B pressure P_B (bar)
Time t (s)
(e)
Actuator displacement x_c − x_c,0 (mm)
Time t (s)
(f) I_S
Threshold
i
x_s
P_A
P_B
x_c
Time t (s)
(g) I_P
Threshold
i
x_s
P_A
P_B
x_c
Time t (s)
Safe zone
Threshold violation